

Unlikely Twins

Let's take a look at the salient core specifications of the G80 and R600, with the R580+, (X1950XTX) and G71 (7900GTX) thrown in for reference.

Some (rare!) common ground and a very significant addition over all previous-generation hardware is the *Geometry Shader Unit*, which is present on both cards. Geometry shaders are an extension of vertex shaders; they can work on many vertices (in 3D terms, called a *mesh*), unlike vertex shaders, which work on a single vertex. The unified shaders on the R600 and G80 and all their derivatives can now perform pixel, vertex, or geometry operations. This translates to efficient use of shader units—the load is balanced between pixel, vertex, and geometry, and none of the shader units are left idle. Traditionally, in the case of heavy, pixel-processing frames, the vertex shader units would idle, though in an always-on state (consuming power).

The R600 a.k.a. HD2900XT's 80nm fabrication process propels it to higher clock speeds within acceptable thermal limits as opposed to the 8800GTX/Ultra a.k.a. G80's 90nm process.

With more than double the shader units and a higher clock, not to mention the 512-bit memory interface, the HD2900XT might seem to bury the 8800GTX, but this isn't an apple-to-apple comparison; the two companies' approach designing and implementing DX10 parts is quite different. The 320 vs. 128 shader-unit argument is equally pointless. And as you will see, there is much more at play than simple on-paper specs.

Is 320 Equal To 128?

The more shaders the merrier, is a general thumb rule. This because a single stream processor can do only so much work in a single clock. Pixel operations in particular are complex and multi-threaded, often needing more than one SP working on the same pixel data (called a dependency).

ATI's 320 Stream Processors: Is more necessarily better?

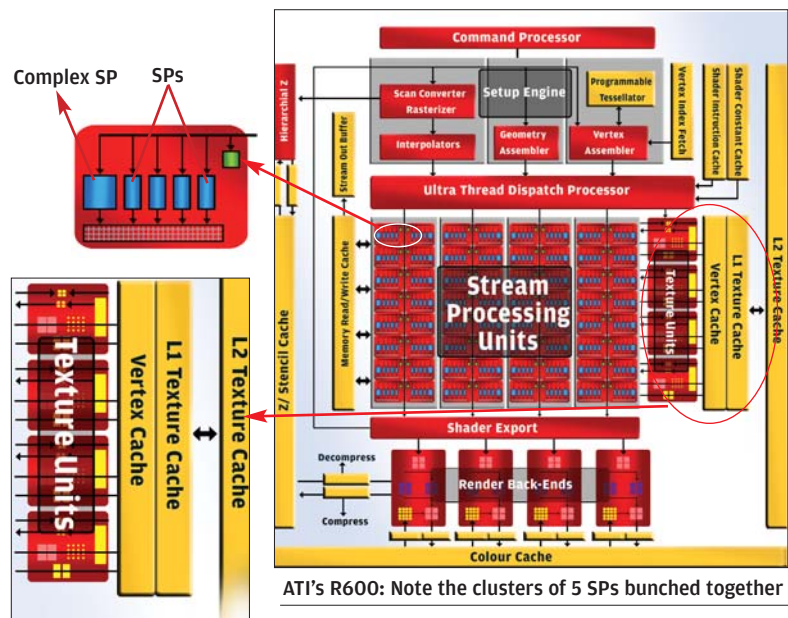
ATI has taken pains to explain that *most* of the Stream Processors (hereon referred to as SPs) on the R600 aren't capable of special function operations, that is, they are simpler than the SPs on the G80. Also, if you look at the diagram alongside, you'll see that five of these SPs are bunched together in a cluster, and only one SP in the block (the large one) can handle either a regular floating point (FP) such as a MADD operation or a special function operation such as SIN (sin), COS (cosine), LOG (logarithm), or EXP (exponent). This SP is called a *special function SP* by ATI. The other four SPs can handle only simpler integer operations such as ADD (add) and MUL (multiply). There is a sixth unit in the SP cluster, which is a branch execution unit responsible for tackling flow control operations like looping, comparisons, and calling subroutines.

Another limiting factor is that each cluster of five SPs can handle only a single thread, be it vertex, pixel, or primitive, in a single cycle. This means essentially that each bunch of five SPs are handling only a single FP operation per cycle, while a single SP in the bunch can do a transcendental operation (read: special operation).

While ATI has gone through pains to ensure that each cluster of 5 SPs is always busy, it is extremely difficult to ensure that each SP in a single cluster is always occupied. Therefore the R600 essentially consists of 64 SPs with 5 ALUs (Arithmetic Logic Units) each, though ATI prefers to refer to the total number of SPs as $64 \times 5 = 320$.

NVIDIA's 128 Stream Processors: Parallelism unparallel?

At the very outset, NVIDIA designed the SPs on the G80 to be independent. No clusters here—just 128 SPs (96 on the 8800GTS) that can perform a single operation per cycle, unless one operation is a simple one. This could be either an FP operation (like a MADD), or a special function (SIN, COS, etc). A single SP can complete a MADD and one integer operation like a MUL in a single clock cycle. So we can have 128 threads running in parallel in a given clock cycle. However this doesn't mean 128 discrete operations because vertex and pixel data contains multiple threads. For example, eight SPs may be working on a single thread in a single clock! Keep in mind that the G80 has eight SIMD blocks each containing a cluster of 16 SPs (see diagram on page xxx).



ATI's R600: Note the clusters of 5 SPs bunched together

Before getting fed to the SPs, each vertex operation is divided into 16 threads, while pixel and geometry operations are divided into blocks of 32 (16 threads are executed over two clocks).

Which is better, 320 or 128?

Each SP on the G80 can perform two simultaneously-issued scalar operations per cycle, say a MADD + MUL. This is in contrast to the R600, where only one SP in a cluster of five can handle such an operation. So the G80 can handle twice the number of threads (128 vs. 64) in parallel.

However, due to the fact that each thread is worked upon by 5x the SPs on the R600, more work is *potentially* getting done on each of the 64 threads than on the 128 threads on the G80. In the *best case scenario*, the R600 can execute five parallel operations on each thread. This is, of course, assuming no dependencies among any of